

Large-scale microbiome profiling studies: opportunities and challenges

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Data availability & standardization

- ✓ Increasing sample sizes & statistical power
- ✓ Comprehensive population coverage & subject stratification
- ✓ Individuality in the larger context

HITChip Atlas:

- > 10,000 samples
- > 5,000 subjects
- > 1,000 phylotypes
- versatile geographical coverage
- some metabolomic profiles etc.
- comprehensive metadata
- intervention studies & long follow-ups (10+ years)
- reproducible & reliable down to 0.1% relative abundance; deep phylotype level profiling
- open data: Lahti et al. Nature Comm. 2014

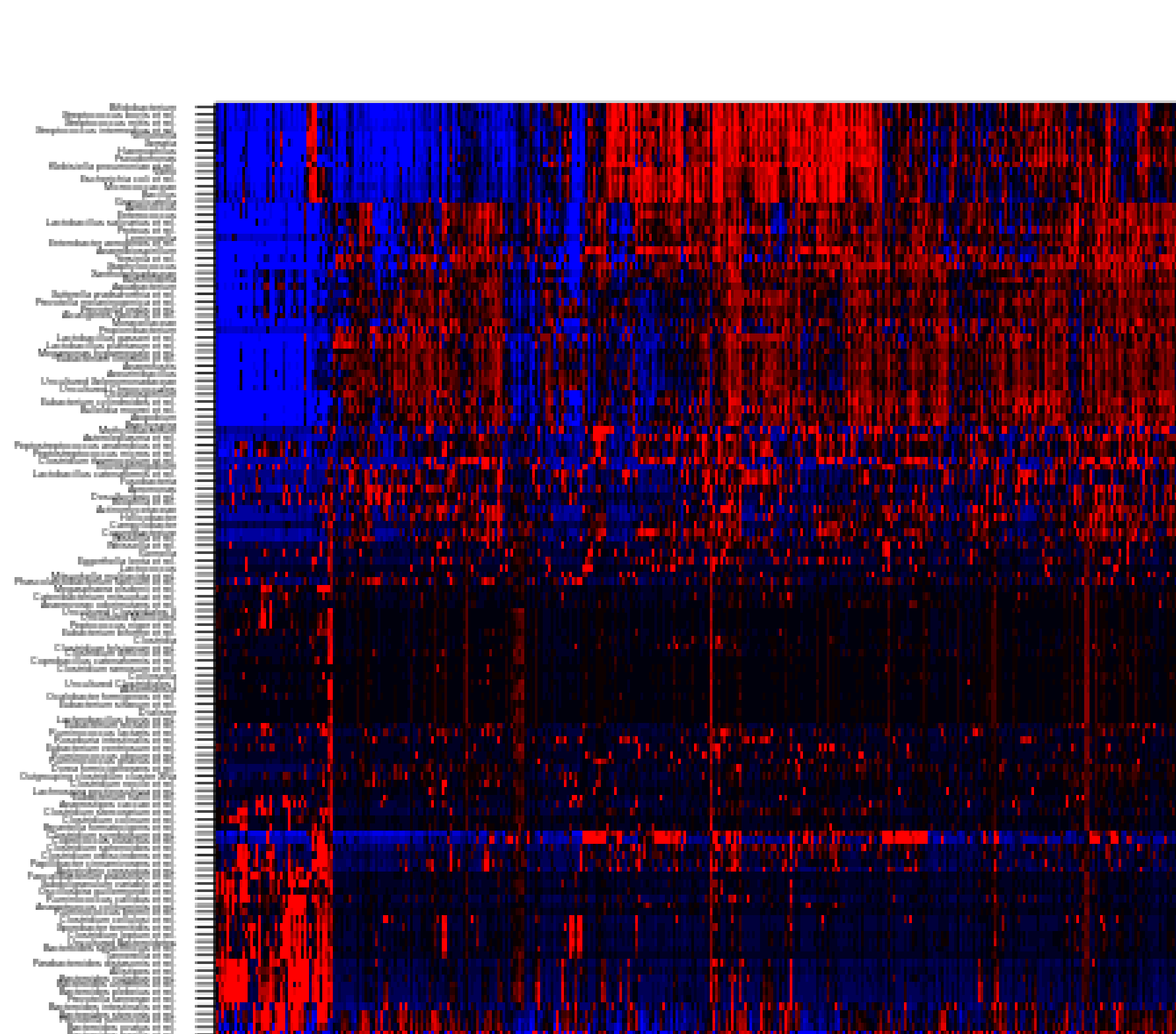


Flemish Gut Flora:

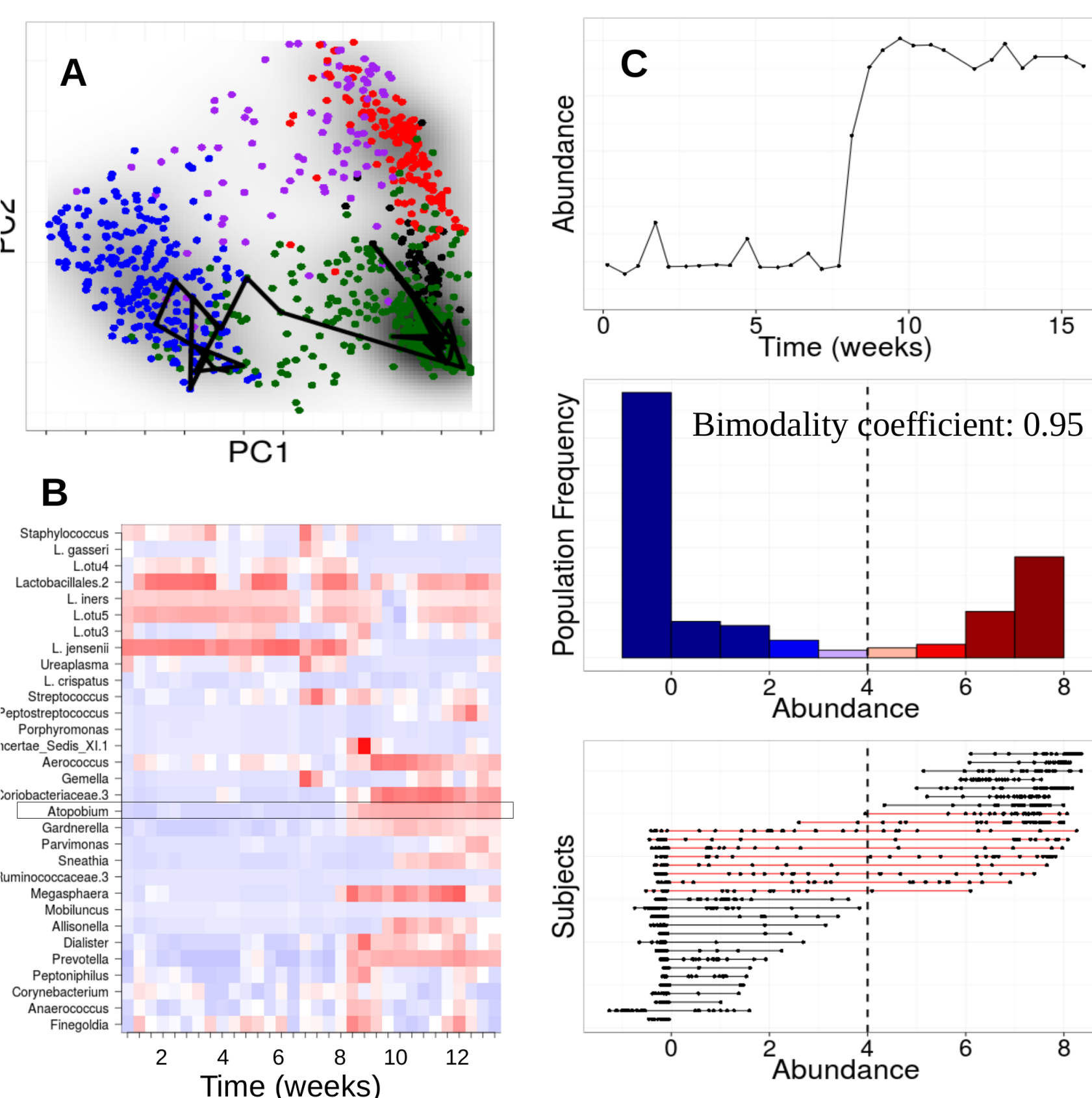
- > 5,000 samples
- > 1,000 subjects
- > 7,000 OTUs
- normal Flemish population
- metagenomic profiling (subset)
- clinical metadata & questionnaires
- longitudinal analyses
- standardized sample collection & processing
- open data: Falony et al. Science 2016



Cross-sectional vs. longitudinal analyses



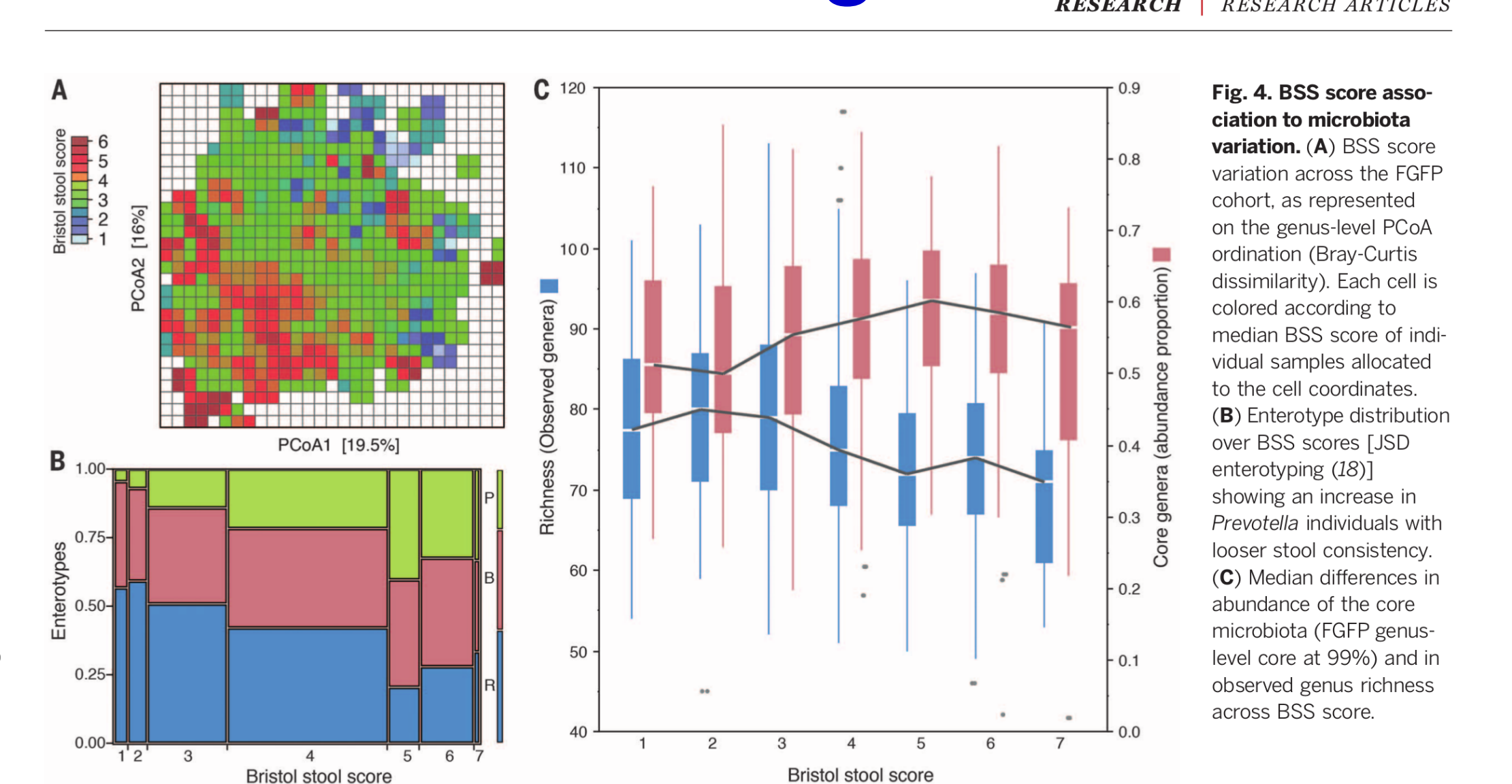
HITChip Atlas: 10,000+ samples;
Data: Lahti et al. 2014.



Vaginal microbiota community dynamics;
Data: Gajer et al. 2012 Figure: Faust et al. 2015

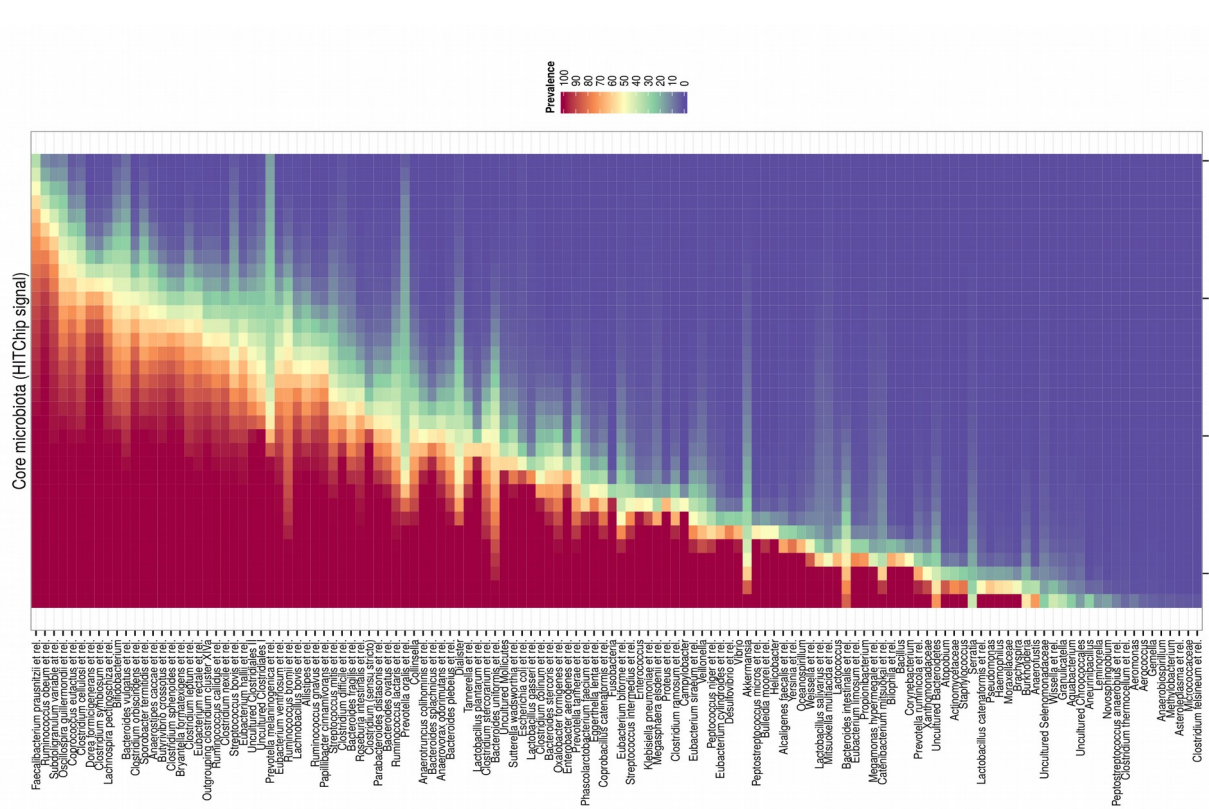
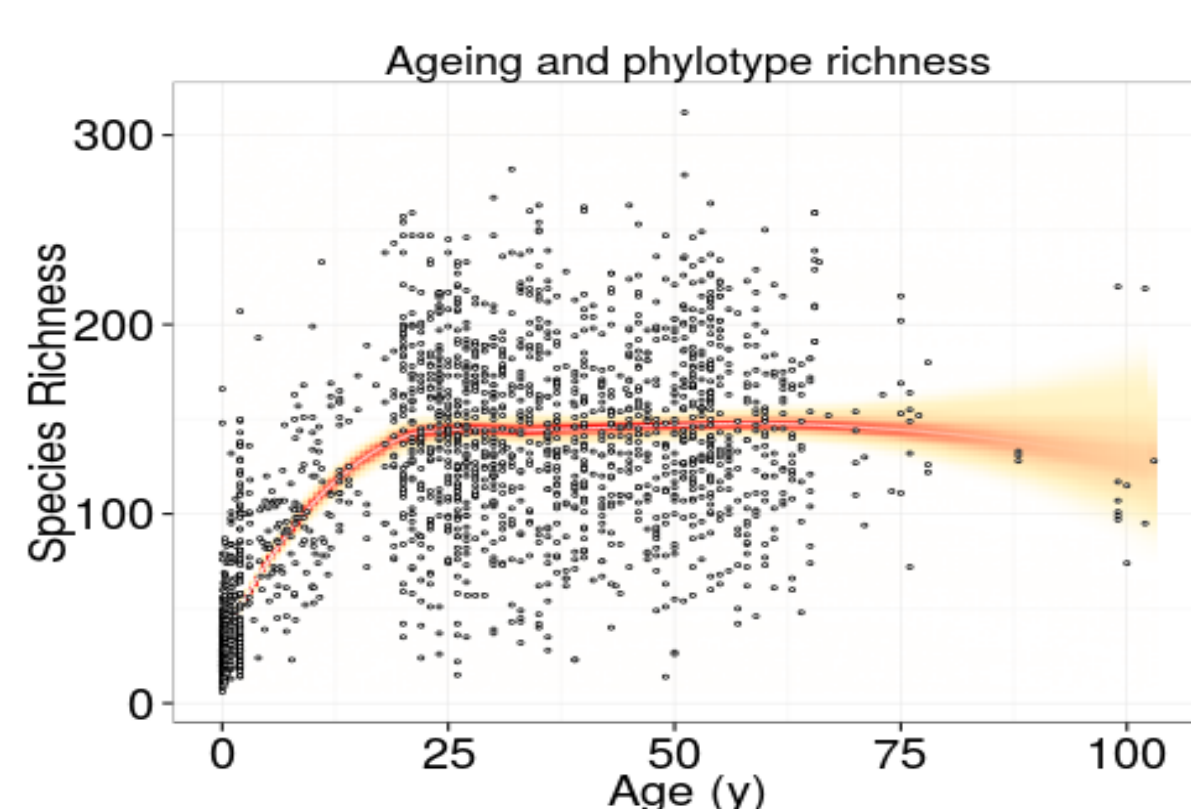
Custom preprocessing,
analysis & visualization tools
<http://microbiome.github.io>

Confounding factors



Stool consistency is a major confounder (Falony et al. 2016)

High individuality

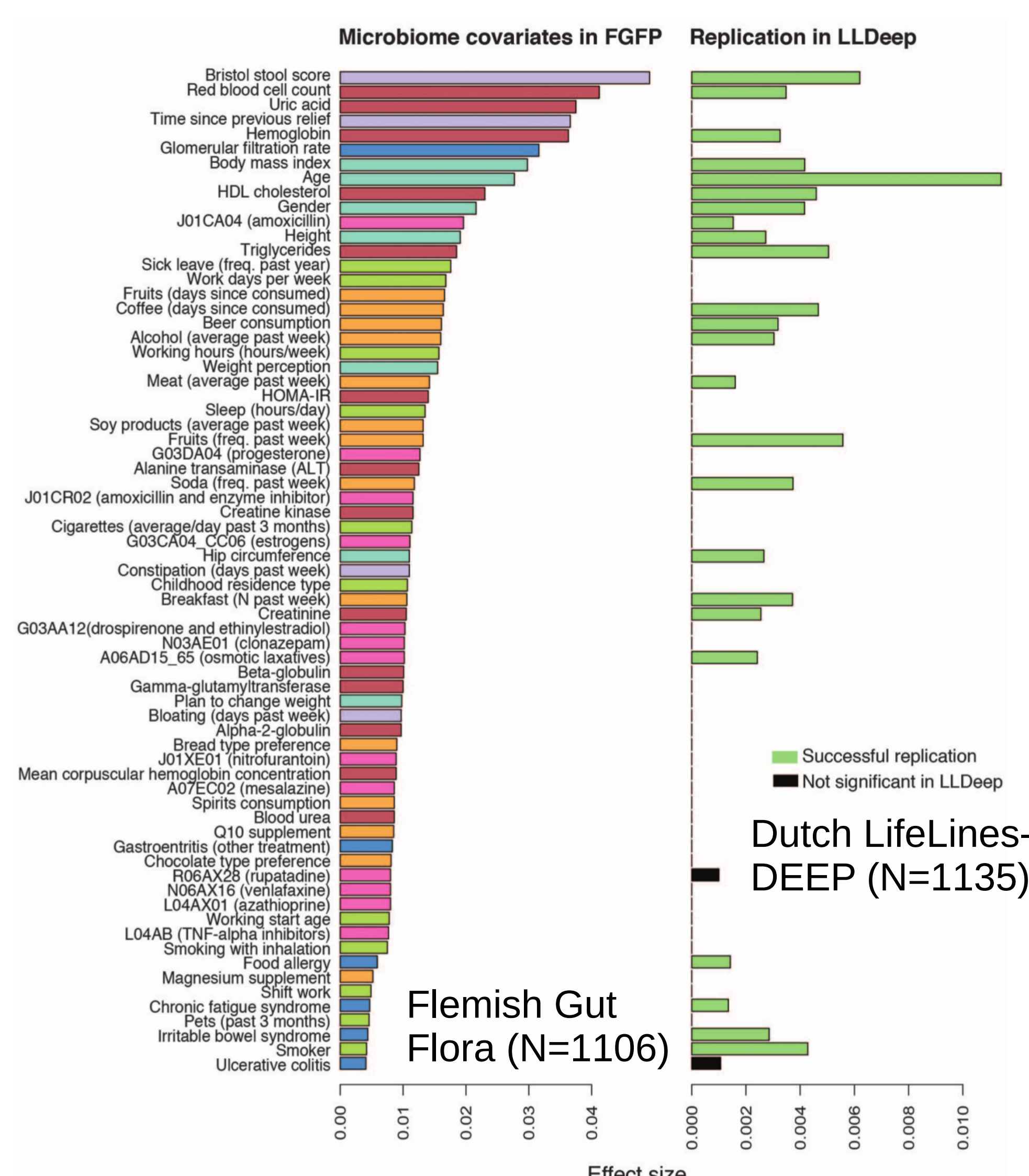
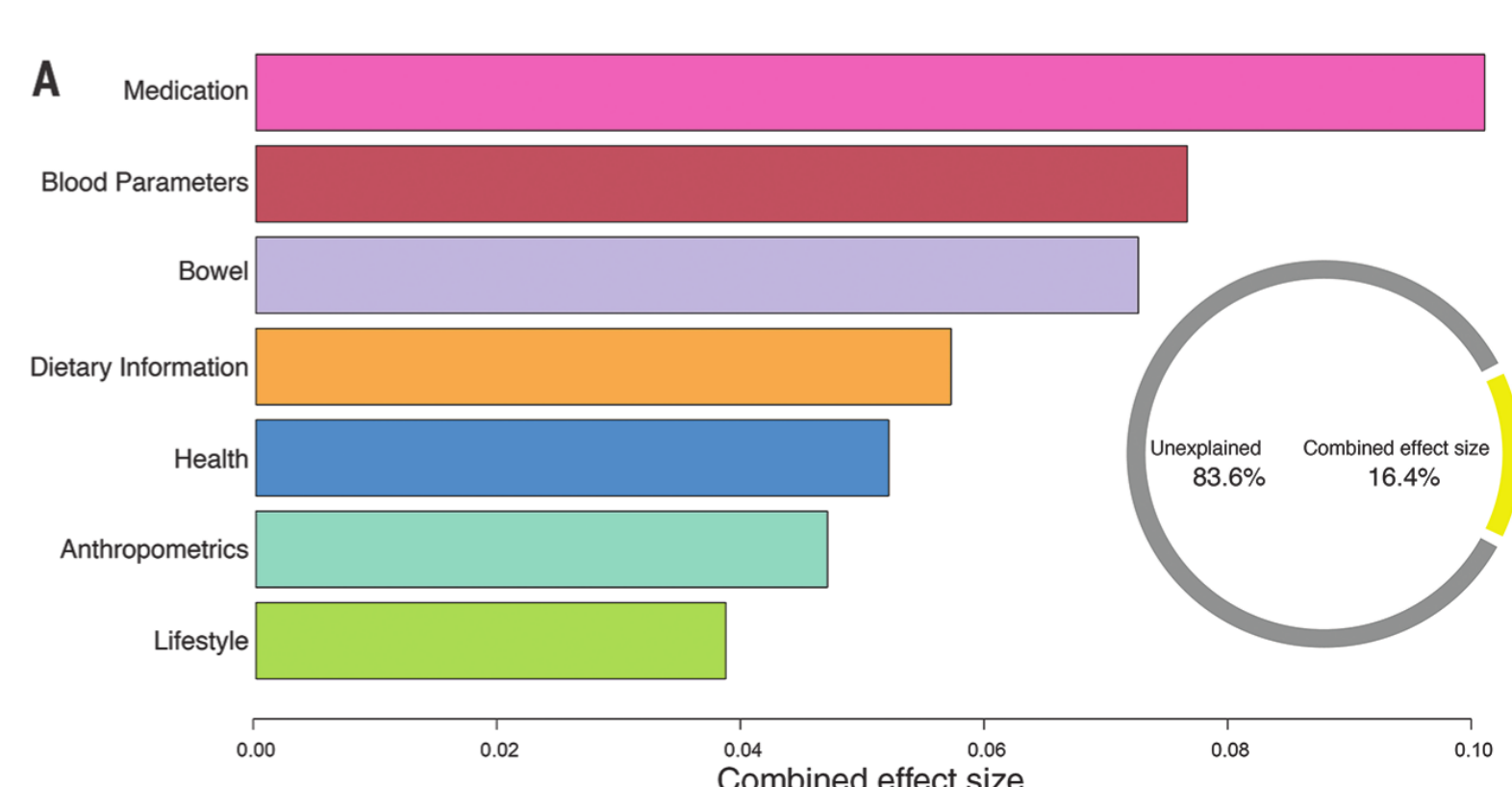


Core microbiota: only few species prevalent at a high abundance
(Data: HITChip Atlas, Lahti et al. 2014)

Complexity

Total explained variation 16.4%:
Proposed disease marker genera associated to host covariates and medication - inclusion in study design is essential! (Flemish Gut Flora; Falony et al. 2016.)

92% replication rate !



References

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- Fat, Fiber and Cancer Risk in African, Americans and Rural Africans. O'Keefe et al. *Nat. Comm.* 6:6342, 2015
- Impact of diet and individual variation on intestinal microbiota composition and fermentation products in obese men. Salonen et al. *ISME Journal* 8:2218-30, 2014
- Tipping elements in the human intestinal ecosystem. Lahti et al. *Nat. Comm.* 5:4344, 2014
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